

MANAGEMENT IACC (413) HND 1 BUSINESS ADMINISTRATION

MEANING OF FINANCIAL MANAGEMENT

Financial management means the management and control of money and money related operations within the businesses. Companies have finance departments that are responsible for these functions. The executive in charge of the finance department is the company's Chief Financial Officer, abbreviated CFO.

The term "financial management" refers to the things that the CFO and the finance department do. These activities include among other things, keeping records, paying employees and suppliers, receiving customer's payment, borrowing money, purchasing assets, selling stocks and paying dividend.

Financial planning ensures that enough funding is available at the right time to meet the needs of the organization for short, medium- and long-term capital while financial control is the process of comparing data on actual performance with expected performance

Financial management refers to that aspect of management which involves planning and controlling the financial affairs of an organization, to ensure that the organization achieves its objectives (particularly its financial objectives). It is basically about the identification of the possible strategies capable of maximizing an organization's market value which involves the allocation of scarce resources among competing opportunities.

It's important to notice that accounting is included in this broad definition of finance and the accounting function is usually part of the finance department reporting the CFO

This involves decisions about:

- ❖ how much finance the business needs for its day-to-day operations and for longer-term investment projects;
- ❖ where the finance should be obtained from (debt or equity) and whether it should be long term (long-term debt or equity) or short term (trade suppliers and bank overdrafts);
- ❖ how to manage short term cash surpluses and short-term cash deficits;
- ❖ amounts to be paid out as dividends; and
- ❖ How to protect the organization against financial risks.

The Corporate Objective of Financial Management

The objective of financial management can be categorized into financial and non-financial with the financial objective considered to be the basic objectives.

- Financial objectives of financial management include;
- i. profit maximization and
 - ii. wealth maximization

Profit Maximization: Traditionally, a business has been considered as an economic institution, as such, it has developed profit as a common and unique measure of efficiency and performance. It is therefore, rational to assume profit maximization as natural business objectives.

Wealth Maximization: This simply means maximizing the net present value of a course of action to shareholders (net present value of a course of action is the difference between the present value of benefits and cost associated).

This objective considers the timing and risk of expected returns by selecting an appropriate discount rate and uses cash flows which are objective for all measurements. The wealth created by a company through its actions is reflected in the market value of the company's shares which is represented by the share price (serving as a performance indicator)

Non - Financial Objectives

- i. welfare of employees; is an attempt to provide good working conditions, training and career development and good pension plan
- ii. Welfare of management; must be ensured as managers will often take decisions to improve their positions at the expense of shareholders
- iii. Welfare of society as a whole; the company's corporate responsibility towards the wellbeing of the society
- iv. Fulfillment of responsibility towards customer and suppliers; fair and honest dealing with Customers providing products and services of their desired quality and also meeting payments and other obligations of suppliers.

The Three Core Decision Area in Financial Management

There are three core decision areas in Financial Management viz:

- ❖ Investment decision
- ❖ Financing decision
- ❖ Dividend decision

1. **Investment decisions.** This involves identification of viable projects i.e. it deals with the appraisal of projects using various techniques so as to determine those that are viable.
2. **Financing decision.** This involves the identification of the appropriate source(s) of finance that would be used to finance the project using various techniques so as to determine those that are viable. Here consideration is usually given to:
 - a. The cost associated with each source
 - b. The characteristic of each source
 - c. The level of risk of the project under consideration
 - d. Pattern of cash flows from the project

- e. Availability of each source
 - f. Taxation and
 - g. The amount involved before the final selection is made
3. **Dividend decision** – here attention is focused on the compensation required by the providers of funds i.e.
- a. Dividend for shareholders- Dividend and the rate of it has to be decided.
 - b. Retained profits- that will be plough back to finance growth and expansion in the company.

THE FINANCE MANAGER

The finance manager is the key personnel who is responsible or the day to day financial services and record keeping of the organization.

He must have a proper understanding of those aspects of legislation which impact upon corporate organizations. Such legislation includes Companies Acts, health and safety regulations, law relating to customer protection rights to contract and agency, employment law and laws relating to protection of the environment.

A finance manager will need to familiarize himself with taxation law as well as established the reasons why all aspect of Business Taxation is relevant to financial strategy.

The main functions of finance manager

They are summarized as follows:

- i. Investment Decision
- ii. Financing Decision
- iii. Dividend Decision
- iv. Liquidity management
- v. Tax management
- vi. Risk management
- vii. Asset management

SOURCES OF FINANCE

Earlier, we identified three broad decisions area in financial management namely: investment decision, financing decision and dividend decision. Investment decisions involve identification of the various investment opportunities and selection based on fund available. Under the financing decision, attention is given to the different sources of funds, the costs associated with each source and the availability of each source.

In this discussion attention is directed to the financing decision with special reference to sources of finance

- 1. Short-term sources – those repayable within one year
- 2. Medium-term sources – those repayable within 1 – 3 years or sometimes 1 – 5 years

3. Long-term sources -- available for 5 years or long

Sources of Short-Term Funds

Short-term funding is used to finance some current assets. (In some cases, companies operate with current liabilities in excess of current assets, but this is unusual.) Most of the usual sources of short-term are:

- Bank credit (bank overdraft)
- Commercial papers
- Trade credits
- Debt factoring
- Bill discounting
- Accruals

Bank credit (bank overdraft)

Commercial Banks sometimes allow their customers to overdrawn their account up to a certain limit, overdraft interest is charged on the day to day overdrawn position, bank overdrafts are usually available for up to one year, but can be rolled over. They are usually unsecured, but under conditions, bank may require collaterals. Additionally, banks may evaluate the credit worthiness of customers through financial statements and cash flow projections before granting the facility.

Commercial Paper

This is an instrument use by large companies to raise short-term funds from the money market. it is usually issued on behalf of the company by issuing house (normally a merchant bank). The issuing house does not guarantee the notes but assist in finding investor to buy them. The investor effectively lends directly to the company issuing the note. The issuing house charges a commission for the service. Commercial paper normally carries a stated coupon rate, and the maturity dates between 30 and 270 days.

Trade Credits

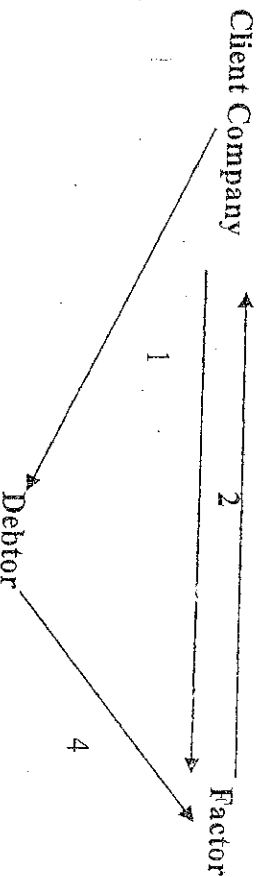
The credit from suppliers is a major source of finance, especially to small companies. trade discount has the advantage of being readily available without formal arrangement. It can be rolled over and therefore will become a continuous source of finance. However, a company should honour its credit arrangements and pay its suppliers on time at the end of the agreed credit period. It is inappropriate for a company to increase the amount of its trade payable by taking excess credit and making payments late.

Debt factoring

This is the raising of funds on the security of the company's debt so that the cash is received earlier than if the company waited for the debtors to pay. It could be service factoring where the

factor buys the company's invoiced debt thereby taking over debt collection and debtor accounting. The procedure involves the company selling goods to the debtor and selling the debt to the factor who may make an initial payment to the company of about 90% of the debt total value while the factor receives a commission for the collection of the debt from the debtors. A factoring company provides immediate cash and also takes care of the collection of the debts as well as the sales administration of debts for a fee.

3



Key: 1 Client sells goods to debtor

2 client sells debt to factor

3 Factor makes payment to client

4 Debtor makes payment to Factor

Bills Discounting

A bill of exchange is normally prepared by the supplier of goods (Creditor) for endorsement/acceptance by the customer (debtor). This is commonly with export sales. The supplier (seller) can obtain immediate cash after the goods have been dispatched by discounting the bill with the bank/discount house.

Accruals

Deferment of tax payments and wages are the commonest examples in this method. Tax laws allows that tax liabilities should be due for payment after one year. Also, employees work for a period of one month before receiving their pay. These form interest -free sources of short-term finance.

Medium Term Sources

The main sources of medium – term funds are

1. Bank Term Loan
2. Venture Capital
3. Equipment Leasing
4. Sale and lease back
5. Hire purchase
6. Mortgage

Bank Term Loan

This is similar to a bank overdraft but carries a higher interest and charge and covers a longer period. A bank loan requires a higher collateral security and is repayable after a specific period of time.

Venture Capital

This is a significant early stage financing of new and young enterprises seeking to grow rapidly. It involves management of the client's enterprises. **Venture capital** refers to the commitment of capital for the formation and setting up of small-scale enterprises specializing in new ideas or new technologies with growth potential.

Equipment Financing

This is a financial arrangement to finance the purchase of an asset through a finance company or a leasing company or a bank. There are two type of lease namely finance and operating lease.

- **Finance Lease**

This is where the risk and benefits of ownership have been substantially transferred to the lessee

- **Operating Lease**

A lease where the risk and benefits of ownership remains with the lessor

Sales and Lease Back

This is a situation where an asset previously owned by a company is disposed of and immediately repossessed through a leasing contract.

Hire Purchase

This is an arrangement under which the hirer, in return for the use of an asset undertakes to make periodic payments to the owner of the asset. He is expected to assume ownership of the asset after the payment of the last installment

Long-Term Sources

The main sources of long-term sources are.

1. Equity Capital
2. Preferential Share capital
3. Debenture Stock Capital

Equity Capital / Ordinary Share Capital

The traditional form of capital is equity capital. The holders of this capital are the owners of the business. They therefore have a general preemptive right to anything of value that the company

may wish to distribute as well as the ultimate control of the company's affairs. They bear a huge portion of the entire risks associated with the company; hence they expect a higher rate of return than most other providers of finance.

Other features are; that the expect and are entitled to a share of the profits of the company in the form of dividends, subject to the recommendation of the directors and after all prior claims have been met. The ordinary shareholders have voting powers of right attached to their investment.

Ordinary shares could take the form of **preferred, deferred or founders'** ordinary shares. **Preferred ordinary shares** usually receive a fixed rate of dividend, before the other ordinary shareholders. They may also be entitled to a further share of profit after their fixed entitlement (dividend).

Deferred ordinary shares are usually residual recipients after all claims including preferred ordinary shareholders have been settled. Deferred shares could be given to the sellers (owners) of a company. These serve as deferred payment for the purchase of the company held back until enough profits emerge. These types of deferred ordinary shares are called founders' share.

Debenture Stock

These are loans of a long-term nature. These could be secured or unsecured. Debenture represents the document acknowledges the indebtedness to the company. In practice the term "debentures" may be restricted to secured loans. The main features.

1. They are not entitled to voting rights
2. They are fixed interest securities entitled to annual interest payment
3. The interest elements are tax deductible
4. They could be redeemable, irredeemable or convertible
5. The principal amounts are usually secured on the assets of the company and could have
 - a) Floating charge or
 - b) Fixed charge or
 - c) A combination of (a) and (b)

Preference Share

Preference shares are usually a more expensive source of finance than debenture stock. This is because debentures are less risky and usually have tax shield (benefit). Other feature of preference shares is that they are not entitled to any voting rights normally, and their interest in the company is represented by dividend payment and principal repayment.

Preference shares could be preferred or deferred, cumulative, participating, redeemable. Cumulative preference shares would have their dividends income accumulated and paid at future dates if the company has liquidity problems.

Participating preference share, shareholders are entitled to a fixed dividend income per year (this may cumulative) plus a further share in any other profits. In some cases, this further share could

be after the ordinary shareholders have been paid a certain dividend. Preferred and deferred preference stock have a characteristic similar to preferred and deferred ordinary shares.

CORPORATE GOVERNANCE

Corporate governance can be defined as the system by which companies are directed and controlled. It is also viewed as the process, structures and relationship through which the Board of Directors oversees what the executives do to achieve the objectives of the organization.

It also means the system by which companies are directed and managed in the best interest of the owner and investors. It refers to the roles of board of directors, executive and non-executives, shareholders' right and to other actions taken by shareholders to influence corporate decisions. It is term that covers the entire general mechanisms by which management are led to act in the best interest of the company's owners.

Essence of Corporate Governance

- a) Corporate Governance aims at promoting a culture in which Directors will give priority to ethical pursuit in the shareholders' best interest
- b) Corporate Governance allows a review of audit regulations, corporate disclosure framework and shareholder participation to improve the accountability and transparency of the companies, compliance to statutory regulation, best ethical practice, consumer protection and so on.
- c) It ensures that the audit committee assists the Board of Directors in managing the accuracy and integrity of the financial statement of companies, ensuring compliance with legal and regulatory requirements, and the efficiency of the company's internal audit functions.
- d) It ensures the credibility of companies and the existence of a managerial system that promotes creative and progressive entrepreneurship.
- e) It assists in the preventing the exploitation of investors by managers
- f) Good Corporate Governance prevent fraudulent practices through the mechanism designed by the Board of Directors and Management
- g) Corporate Governance ensures that the providers of finance to companies have their rewards or returns.
- h) Corporate Governance helps in maximizing the corporate value by enhancing transparency and efficiency.

Principles of Corporate Governance

Good corporate governance should be designated in line with the circumstances surrounding each corporation and continuously reviewed according to changing circumstances of time. However, for guidance among companies, which intend to compete international and raise

managerial efficiency, the following could serve as the basic principles of corporate governance to be adopted.

- i. Lay solid foundation or management oversight
- ii. Structure the board to add value
- iii. Promote ethical and responsible decision making
- iv. Safeguarding the integrity of financial reporting
- v. Making continuous, timely and balanced disclosure to Stock Exchange
- vi. Respect the rights of shareholders
- vii. Encourage enhanced performance evaluation
- viii. Remunerate fairly and responsibly
- ix. Recognize the legitimate interest of shareholders

TREASURY FUNCTION

The treasury department is responsible for making sure that cash is available in the right amounts, at the right time and in the right place. Its roles include

- ❖ Produce regular cash flow forecasts to predict surpluses and short falls
- ❖ Arrange short term borrowing and investment when necessary
- ❖ Deal with entity's banks
- ❖ Finance the business on a day to day basis
- ❖ Advise senior managers on long term financing requirements
- ❖ Advise on foreign exchange transactions
- ❖ Risk management such as foreign exchange or currency risk, interest rate risk, credit risk or market risk. (credit risk is a risk that a debt will not be paid when due while market risk is the risk of adverse movement in the market price of assets)

CAPITAL MARKET

A capital market is a financial market in which long term capital is bought and sold. A stock market is a capital market for stock and shares. A financial market where short-term capital is borrowed and lent is a money market.

The role of a stock market for companies is, to

- ❖ Bring companies and investors together
 - investors put new long-term funds into a company
 - a company can raise capital for investment in the business
- ❖ Provide a second hand market for stocks and shares, whereby
 - investors can sell stocks or share if and when to do so
 - other investors can acquire existing stocks or share in a company

Further roles of a stock market are to

- Enable the owners of a company to realise some of their investment by bringing their hitherto private company to the market
- Allow companies to take over other companies by issuing new share as the purchase consideration. Stock market companies can therefore use their market status to finance expansion through acquisition.

Not all companies seek a stock market quotation for their share. Reasons for remaining private' might be

- ▶ Insufficient size
- ▶ Insufficient past trading record
- ▶ No requirement for extra capital or rapid growth: adequate private funding
- ▶ To avoid pressure of stock market investors' expectations for earnings and dividend growth
- ▶ To avoid widespread share ownership; to retain ownership in the hands of a few individuals, to avoid takeover threats.

MATHEMATICS OF FINANCE

The Time Value of Money

One of the basic principles of finance is that a sum of money today is worth more than the same sum in the future. If offered a choice between receiving ₦ 10,000 today or in 1 years' time a person would choose today.

A sum today can be invested to earn a return. This alone makes it worth more than the same sum in the future. This is referred to as the time value of money.

The impact of time value can be estimated using one of the two methods:

1. compounding which estimates future cash flows that will arise as a result of investing an amount today at a given rate of interest for a given period
 - An amount invested today is multiplied by a compound factor to give the amount of cash expected at a specific time in the future assuming a given rate.
2. Discounting which estimates the present-day equivalent (present value which is usually abbreviated to **PV**) of a future cash flow at a specified time in the future at a given rate of interest.
 - An amount expected at a specified time in the future is multiplied by a discount factor to give the present value of that amount at a given rate of interest.
 - The discount factor is the inverse of a compound factor for the same period and interest rate. Therefore, multiplying by a discount factor is the same as dividing by a compound factor
 - Discounting is the reverse of compounding

Example 1

Musa places ₦ 25,000 on deposit with key ball bank. Interest at 12% p.a is paid at six monthly intervals direct into his current account.

How much interest will he received over a four-year period?

Solution

12% p.a equivalent to 6% half year. Therefore, the amount of interest received every half year is:
 $6\% \times ₦ 25,000 = ₦ 1,500$ and the total amount received over four years is:
 $₦ 1,500 \times 2 \times 4 = ₦ 12,000$

Saying

Interest = ₦ 25,000 $\times$ 12% $\times$ 4 years = ₦ 12,000

Compound interest formula

Compounding involves the payment of interest on earlier interests and can be calculated using the formula below:

$$A = P(1+r)^n$$

Where:

A	=	final amount
P	=	initial investment
r	=	interest rate per period
n	=	number of periods

Example 2

Amara Chukwuana is to invest ₦ 100,000 in a bank for 3 years at an annual interest rate of 18%. Required

- What is the future sum from this investment?
- What is the interest element?

Solution

$$A = P(1+r)^n$$

$$\begin{aligned} \text{a) } A &= 100,000(1.18)^3 \\ &= ₦ 164,303.20 \end{aligned}$$

- Calculation of interest receivable

₦

Future sum	164,303.20
Less principal sum	<u>100,000.00</u>
Interest element	<u>64,303.20</u>

Example 3

Yaks deposits ₦ 100,000 in bank for a period of five years. Interest is compounded at the following rates

Year	1	2	3	4	5
Rate	10%	10%	12%	15%	15%

Required:

- What is the balance in the account at the end of the project?
- What is the equivalent annual compound rate earned on the investment?

Solution

a)	Original principal	₦ 100,000
Year 1	interest at 10%	<u>10,000</u>
		110,000
Year 2	interest at 10%	<u>11,000</u>
		121,000
Year 3	interest at 12%	<u>14,520</u>
		135,520
Year 4	interest at 15%	<u>20,328</u>
		155,848
Year 5	interest at 15%	<u>23,377.20</u>
		<u>179,225.20</u>
	Balance	

The fastest approach is to use the compound interest formula

$$A = ₦ 100,000(1.1)^2(1.12)(1.15)^2 = ₦ 179,225.20$$

Discounting

Formula: Discount factor

$$\text{Discount factor} = \frac{1}{(1+r)^n}$$

Where:

$$\begin{aligned} r &= \text{the period interest rate} \\ n &= \text{number of periods.} \end{aligned}$$

Example: Discounting

A person expects to receive ₦ 13,310 in 3 years. If the person faces an interest rate of 10%. What is the present value of this amount?

$$\text{Present value} = \text{Future cash flow} \times \frac{1}{(1+r)^n}$$

$$PV = \text{₦}13,310 \times \frac{1}{(1.1)^3} = 10,000$$

$$PV = \text{₦}10,000$$

Discounting is the reverse of compounding

Class work

1. Calculate the present value of ₦ 60,000 received in 4 years, assuming a cost of capital of 7%.
2. An investor deposit ₦ 45,780 in a bank for four years at 7% p.a. compound interest. What is the value of the account at the end of the period?

SINKING FUNDS

A person may save a constant annual amount to produce a required amount at a specific point in time in the future. This is known as a sinking fund.

Example Sinking fund

A man wishes to invest equal annual amounts so that he accumulates ₦ 5,000,000 by the end of 10 years. The annual interest rate available for investment is 6%. What equal annual amounts should he set aside.

Solution

Step 1 Calculate the present value of the amount required in 10 years.

$$PV = 5,000,000 \times \frac{1}{(1.06)^{10}} = 2,791,974$$

Step 2: Calculate the equivalent annual cash flows that result in this present value

Present value	₦
Divide by the 10 years, 6% annuity factor	2,791,974
Annual amount to set aside	<u>7.36</u>
	<u>379,344</u>

If the man invests ₦379,344 for 10 years at 6%, he will accumulate to ₦ 5,000,000

Or

Another version of the annuity factor formula

$$\text{Sinking fund factor} = \frac{(1+r)^n - 1}{r}$$

where

r = interest rate, as a proportion

n = number of time periods

Step 1: Estimate the sinking fund factor

$$\frac{(1 + 0.06)^{10} - 1}{0.06} = 13.181$$

Step 2: Calculate the annual amount to be set aside by dividing the target amount by the sinking fund factor

Present value	₦
Divide by the 10 years, 6% sinking fund factor	5,000,000
Annual amount to set aside	<u>13.181</u>
	<u>376,344</u>

CAPITAL BUDGETING

Introduction

This section concentrates upon the techniques available to determine whether or not news investments or projects should be undertaken. The techniques can be applied to any type of investment (made by individuals or companies where the financial implications can be identified). They are, however examined exclusively in relation to corporate investments made with the objective of maximizing their benefit to ordinary shareholders, i.e., maximizing shareholders wealth. An investment simply means something which involves an initial capital outlay and which produces future cash inflows.

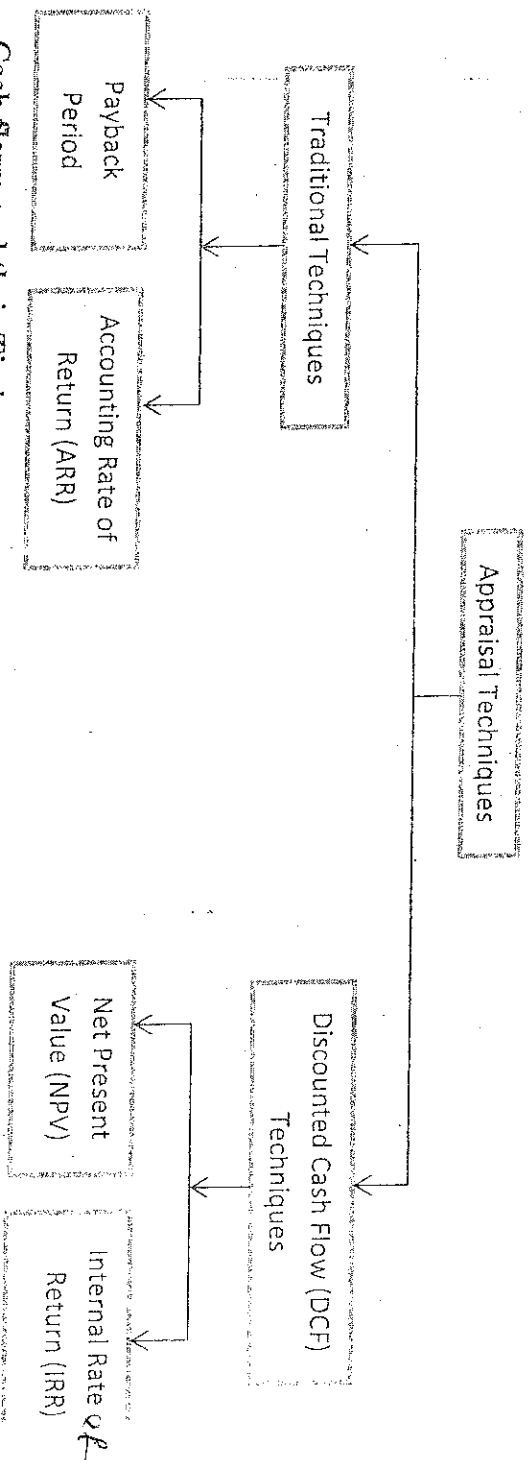
The money companies spend in the normal course of business can be divided into two categories funds are expanded on an everyday basis by buying inventories, pay expenses and companies' employees these expenses can be thought of as short term in that they support daily activity. In addition to short term expenditures, firms spend large sum on special project from time to time. For examples, machines normally wear out and need to be replaced. The replacement expenditure tends to be relatively large but they are infrequent. New business ventures provide a second example because they generally require initial spending to get started. Startup amounts are usually large, but the opportunities don't come along often. Spending on things like these is long term because the projects involved tend to last for long periods. As a general rule, money spend on long-term projects is called capital.

A capital budgeting decision may be defined as the firm's decision to invest its current funds most efficiently in the long term assets with anticipation of an expected flow of benefits over a series of years.

The investment decision of a firm (expansion, acquisition, modernization and replacement of the long term assets) is generally known as the capital budgeting decision.

CHARACTERISTICS / FEATURES OF INVESTMENT DECISIONS

- The decision involves a large initial outlay
- It involves exchange of current funds for future benefits.
- It is a risky decision
- The decision is irreversible
- The funds are invested in long term assets.
- The future benefits will accrue to the firm over a series of year



Cash flows and their Timing

The following sign conventions apply to all investments:

- Cash outflows or expenditure are represented by negative figures
- Cash inflows or income are represented by positive figures

The timing of cash flows is very important in investment appraisals, for convenience an annual time scale is normally used where:

Year 0: represents the date of making the investment

Year 1 represents the date 1 year after the initial investment. It is the end of the first year in the life of the investment and also it is the beginning of the second year

Year 2 represents the date 1 the end of the second year and the beginning of the third year of an Investments.

INVESTMENT APPRAISAL TECHNIQUES

There are two major classifications of investment appraisal techniques

- ❖ **Traditional or Non-discounted cash flow technique** – which refers to a method of appraisal which does not consider the time value of money, and they include
 - payback period method and
 - Accounting rate of return.
- ❖ **Modern or Discounted cash flow technique**

The principles and decision rules are the same as in the normal payback period method, the only difference, is that the cash flows to be used are discounted at the given or appropriate cost of capital. Therefore, this version of payback technique, will not suffer from the disadvantage of discountance of time value concerned. It includes the following

- Net Present Value (NPV)
- Internal Rate of Returns or DCF yield

Payback Period Method

The payback period is the time taken in years for a project to recover its initial investment. The shorter the payback period the more valuable the investment is. If the project generates constant annual cash inflows, the payback period can be computed by dividing the cash outlay by the annual cash inflows, that is,

$$\text{Payback} = \frac{\text{Initial Outlay}}{\text{Annual cash inflow}}$$

It is emphasized however, that the payback uses cash flow from item and not financial accounting profit. Thus items like depreciation and other non-financial accounting item are excluded

Example 1

Project costs ₦40,000 to initiate and it is expected to have a useful life of ^{Ten} years. It will generate annual profit of ₦4,000 after charging all expenses, including depreciation at 10 % p. a on cost. Calculate the payback period.

Solution

Profit after depreciation	₦ 4,000
Add back depreciation (10% of 40,000)	<u>4,000</u>
Annual cash inflows	<u>8,000</u>

$$\text{Payback} = \frac{\text{₦40,000}}{\text{₦8,000}} = 5 \text{ years}$$

Example 2

Testy limited is a manufacturing outfit, having a project which involves an immediate cash outlay of 200,000. The company estimate that the net cash inflows from the project will be as follows.

Years	cash flows
	₦
1	20,000
2	40,000
3	220,000
4	80,000

Calculate the payback period for the above project

TESTY LIMITED – INVESTMENT APPRAISAL

Year	Cash flows ₦	Cumulative Cash Flows ₦
0	(200,000)	(200,000)
1	20,000	(180,000)
2	40,000	(140,000)
3	220,000	80,000
4	80,000	

$$\text{Payback period } 2 + \frac{80}{220} = 2.36 \text{ years}$$

DECISION RULE

- Using the payback method, accept all projects whose payback periods are shorter than the company's predetermined minimum acceptable payback period.
- If mutually exclusive projects are involved, whereby only one of the projects can be undertaken and others rejected, the rule is to accept with the shortest payback period

ADVANTAGES

- It is simple to calculate and understand
- Of all the methods of capital budgeting, it least exposes the firm to problems of uncertainty, since it focuses on shortness of project to pay back the initial outlay.
- It is a fast screening techniques, especially for firm that have liquidity problems

DISADVANTAGES

- it does incorporate the time value of money, that is, it does not recognize the fact that the value of ₦1 today will be far more than the value of ₦1 in two- or three-years time
- it ignores cash flows after the payback periods
- it does not take into account the risk associated with each project and the attitude of the company to risk

Practical Questions

A company must choose between two investments, Project A and Project B. It cannot undertake both investments. The expected cash flows for each project are:

Year	Project A ₦	Project B ₦
0	(80,000)	(80,000)
1	20,000	60,000
2	36,000	24,000
3	36,000	12,000
4	17,000	-

The company has a policy that the maximum permissible payback period for an investment is three years and if a choice has to be made between two projects, the project with the earlier payback will be chosen.

Required

- i. Calculate the payback period for each project:

ACCOUNTING RATE OF RETURN METHOD (ARR)

This method is based upon accounting profits and not cash flow. It uses financial accounting profits after charging all financial accounting expenses including depreciation. It is best defined as the ratio of average profit after tax to the average investment.

$$\text{ARR} = \frac{\text{Average Annual Accounting Profit}}{\text{Average investment}} \times 100$$

$$\text{Where Average Annual Accounting profit} \Rightarrow \frac{\text{Total profit}}{\text{Number of years}}$$

$$\text{Average investment} \Rightarrow \frac{\text{Initial investment} + \text{Scrap value}}{2}$$

It is also called Return on investment or Return on capital employed.

Example

Samloko Ltd is considering the purchase of a new machine which is expected to save labour on an existing project. The cashflows on associated project are as follows

₦ 000

Initial cost (year 0) 120

Residual value of the machine 20

Annual labour cost savings

Year 1	40
2	40
3	40
4	20
5	20

Calculate the ARR.

Solution

₦ 000

Total profit before depreciation 160

Total depreciation 120 - 20

Total accounting profit 60

$$\text{ARR} = \frac{\text{Average Annual Accounting Profit}}{\text{Average investment}} \times 100$$

$$\text{Average investment} \Rightarrow \frac{\text{Initial investment} + \text{Scrap value}}{2}$$

$$\text{Average investment} \Rightarrow \frac{120,000 + 20,000}{2} = \text{₦ } 70,000$$

$$\text{Average accounting profit} = \frac{60}{5} = \text{₦ } 12,000$$

$$\text{ARR} = \frac{12,000 \times 100}{70,000} = 17.14\%$$

Accept/reject criterion

Usually, a target criterion is set by management, if this is exceeded, then there is prima facie case for undertaking the project

Mr. Millennia recently convinced his friends and relations to grant him a loan of N200,000 which he intends to invest in a farming project. He estimates that the project will yield the following returns annually for next five consecutive years.

Year	₦
1	60,000
2	60,000
3	80,000
4	60,000
5	40,000

There was no scrap value, at the end of the fifth year and the company desires to evaluate the project on the basis of accounting rate of return.

Required:

Provide the accounting rate of return of this project on the assumption that the annual returns are profits after depreciation but before taxation.

$$\text{ARR} = \frac{\text{Average Annual Accounting Profit}}{\text{Average investment}} \times 100$$

$$\frac{60,000 + 60,000 + 80,000 + 60,000 + 40,000}{5} = \text{₦ } 60,000$$

$$\text{Average investment} \Rightarrow \frac{\text{Initial investment} + \text{Scrap value}}{2}$$

$$\text{Average investment} \Rightarrow \frac{200,000 + 0}{2} = 100,000$$

$$\text{Therefore, ARR} = \frac{60,000}{100,000} \times 100 = 60\%$$

Decision Rules:

(a) The rule is to invest in all projects whose accounting rate of return are higher than the company's predetermined minimum acceptable ARR.

(b) Where mutually exclusive projects are concerned, the rule is to accept the project with the highest ARR.

Advantages

- a. it is easy to calculate
- b. unlike the payback period, it makes use of all the accounting
- c. For divisionalised companies, manager would find the technique easier to understand because it is similar to their annual performance evaluation techniques.

Disadvantages

- i. it does not recognize the time value of money
- ii. It does not consider risk and managements attitude towards risk
- iii. It is an average concept and as such will hide the sizes and timing of the individual cash flows.

Practical Question

Samloko Ltd is considering the purchase of a new machine which is expected to save labour on an existing project. The cashflows on associated project are as follows

₦ 000

Initial cost (year 0) 120

Residual value of the machine 20

Annual labour cost savings

Year	1	2	3	4	5
	240	40	40	20	20

Calculate the ARR

MODERN OR DISCOUNTED CASH FLOW TECHNIQUE

The principles and decision rules are the same as in the normal payback period method, the only difference, is that the cash flows to be used are discounted at the given or appropriate cost of capital. Therefore, this version of payback technique, will not suffer from the disadvantage of discountance of time value concerned.

The following two approaches will be discussed

- ❖ Net Present Value (NPV)
- ❖ Internal Rate of Returns or DCF yield

Net Present Value Method:

Net present value (NPV) is the value obtained by discounting all cash outflows of a project by a chosen target rate of return or cost of capital. The NPV therefore compares the present value of all the cash inflows from an investment with the present value of all cash outflows from an investment. The present value of cash inflows minus the present value of cash outflows is the NPV.

- i. If the NPV is positive, it means that the cash inflows from a capital investment will yield a return in excess of the cost of capital; and so the project should be undertaken if the cost of capital is organization's target rate of return.
- ii. If the NPV is negative it means that the cash inflows from a capital investment will yield a return below the cost of capital, and so the project should not be undertaken if the cost of capital is the organization's target rate of return.
- iii. If the NPV is exactly zero, the cash inflow from a capital investment will yield a return which is exactly the same as the cost of capital, and so if the cost of capital is the organization's target rate of return, the project will be only just worth undertaken.

AB Ltd is considering a capital investment, where the estimate cash flows are

Year	Cash flow
0	(30,000)
1	4000
2	16,300
3	10,000
4	18,000

The company cost of capital is 16%. What is the NPV of the project and should it be undertaken

Solution

Year	NCF ₦	DCF @16%	PV ₦
0	(30,000)	1	3,440
1	4,000	0.86	4,187 3446
2	16,300	0.74	6,400 12062
3	10,000	0.64	9,500 6460
4	18,000	0.55	9,900 99
	NPV		<u>1,580</u> 1802

The NPV is positive, if other factors remain the same, the project should be accepted.

Practical Questions

A company is considering whether to invest in a new item of equipment costing 53,000 to make a new product.

The product would have a four-year life, and the estimated cash profits over the four year period are as follows.

Years	N
1	17,000
2	25,000
3	16,000
4	12,000

Calculate the NPV of the project using a discount rate of 11%

INTERNAL RATE OF RETURN (IRR)

The IRR method is another method of investment appraisal using DCF. The internal rate of return a project is the discounted rate of return on the investment

Estimating the IRR of an investment project

To estimate the IRR, you should begin by calculating the NPV of the project at two discount rates

- One of the NPVs should be positive, and the other NPV should be negative.
- Ideally, the NPVs should both be close to zero, for better accuracy in the estimate of the IRR

When the NPV for one discount rate is positive NPV and the NPV for another discount rate is negative, the IRR must be somewhere between these two discount rates.

Calculation of IRR

Determine NPV using the given and cost of capital

Determine another NPV which must be opposite in sign to that calculation in "1" above (if the initial NPV is positive use a higher discount rate to get a negative and vice versa).

Interpolate using the formulae $IRR = \text{lower rate} + \text{NPV of lower rate}$

$$(\text{NPV}_{\text{LOWERRATE}} - \text{NPV}_{\text{HIGHER}})$$

Formula: IRR interpolation formula

$$IRR = LR + \frac{\text{NPV}_{LR}}{\text{NPV}_{LR} - \text{NPV}_{HR}} \times (HR - LR) \%$$

DECISION RULE

Independent projects: Accept all projects whose IRR is greater than the company's cost of capital.

Mutually exclusive projects: Accept the project with the highest IRR.

ADVANTAGES OF IRR

1. It recognizes the time value of money
2. It considers all cash flows over the project's life span
3. It is consistent with the objective of shareholders wealth maximization
4. It is objective as it makes use of cash flow
5. It provides to us a margin of safety in the calculation of a company's cost of capital
6. It is more attractive to divisional managers since they are used to the return approach in evaluations.

DISADVANTAGES OF IRR

1. It is difficult to calculate compared to other methods
2. A project may have multiple rates of return
3. Its decision may conflict with NPV where mutually exclusive projects are being considered under certain situations.

4. It is a trial and error method as it gives a relative profitability
5. The IRR's of different projects cannot be added
6. It may be difficult to estimate the cash flows
7. It does not consider risk or management's attitude towards risk.

Practice questions

1 The following information is about a project.

Year	₦	DCF 15%	Py
0	(53,000)	1.00	(53,000)
1	17,000	0.86	14,626
2	25,000	0.73	18,250
3	16,000	0.63	10,080
4	12,000	0.54	6,448
			<u>68,404</u>
			15,404

This project has an NPV of ₦2,210 at a discount rate of 11%. The CoC is 11% calculate the IRR of the project.

2 The following information is about a project.

Year	₦
0	(65,000)
1	27,000
2	31,000
3	15,000

This project has an NPV of ₦ (1,515) at a discount rate of 8%

Estimate the IRR of the project
Note: If your first NPV is positive to get the second NPV which will be negative increase your cost of capital

CAPITAL RATIONING

The decision rule is for net present value is to accept all project with positive NPV shareholder wealth is maximized if the company undertakes the entire project with positive NPV.

Capital rationing is a situation where there are insufficient funds to undertake all projects with positive NPV. For example

Project	A	B	C	D	E
Outlay	30m	5m	7m	7m	10m
NPV	+2.5m	+0.5m	+0.7m	+0.8m	+1m

Available fund is 40m

Required fund is 47m

Type of Capital Rationing

- ❖ Hard
- ❖ Soft

Hard Capital Rationing: These are capital rationing not within the control of the company. It is often caused by external factors.

Causes of Hard Capital Rationing

- a. Where financial institutions are not willing to lend to the company at whatever rate
- b. Global policies that may affect the country where the company is located

Soft Capital Rationing: These are capital rationing that are within the control of the company. It is often caused by internal factors such as:

- a) Management may be reluctant to issue additional share capital because of concern that this may lead to outsiders gaining control of the business
- b) Management may not want to raise additional debt capital because they do not wish to be committed to large fixed interest payment
- c) Management may want to limit investment to a level that can be financed solely from retained earnings

Different Situations of Capital Rationing

1. Independent project / stand-alone project. These are projects that are to be appraised for acceptance individually or appraised them for rejection individually. The rejection of one of the projects does not lead to rejection of another
2. Mutually dependent project: these are appraised for joint acceptance i.e. the two projects have to be accepted at the same time or you reject both. If you accept project A, you also have to accept project B.
3. Mutually exclusive project: these are project that cannot be accepted at the same time i.e. once you accept project A, you cannot accept project B and vice versa
4. Divisible projects: these are projects that can be undertaken in parts. Where there is no sufficient fund, you can undertake a fraction of the project, for example if you need 40m and fund available is 20m, you can decide to finance 50% of the project.
5. Indivisible projects: These are project that cannot be part financed i.e. fraction cannot be undertaken
6. Single period capital rationing: This is where fund constraint span is within one year. For example, there is fund constraints in year one while in year two there is sufficient fund
7. Multi period capital rationing. This is the opposite of single period capital rationing. This is where there is fund constraints span over multiple period or more than a year.

Step in solving single period capital rationing

- Compare the NPV of each project
- Determine the profitability index (PI) of each project with positive NPV
- Rank the projects on the basis of PI
- Choose the optimal combination of the projects

Profitability Index (PI)

This is actually a benefit/cost analysis of projects, that is, the N receivable per N of expenditure. It can be measured as the ratio of NPV or in some cases, Gross Present Value (GPV) of a project to the outlay required for the project during the year of restriction. The PVs are the contributions while the expenditure or the outlays are the limiting factors. However, two definitions of PI are allowable as follows:

$$(a) \text{ PI} = \frac{\text{NPV of a project} \times 100}{\text{PV of Outlay}} \quad 1$$

Example

Consider the following projects

Project	Required Outlay ₦ 000	NPV at 10% ₦ 000
A	100	20
B	300	35
C	50	16
D	200	25
E	100	30

Available funds ₦300,000

Determine the optimum combination of the project assuming the projects are divisible

Solution

Project	PI = NPV/Outlay	Ranking
A	0.200	3
B	0.117	5
C	0.320	1
D	0.125	4
E	0.300	2

Application of Available Capital

Project	Outlay	NPV
	₦ 000	₦ 000
C	50	16
E	100	30

A	100	20
50/200		
25% of D	$\frac{50}{300} (0.25 \times 25)$	$\frac{6.25}{72.25}$

Recommendation: the company should select projects C,E and A and 25% of D leading to a total NPV of ₦ 72,250,000

Example

Using the same data as in above but assume the project are indivisible

Possible combinations	Total Outlay	Total NPV
	₦000,	₦000
A, D	300	45
A, C, E	250	66 4 optimal
B	300	35
C, D	250	41
D, E	300	55

Recommendation: Select projects A, C and E